

Forecasts and Long-term probabilistic data Assimilative prediction of the effects of Geomagnetic storms (FLAG)

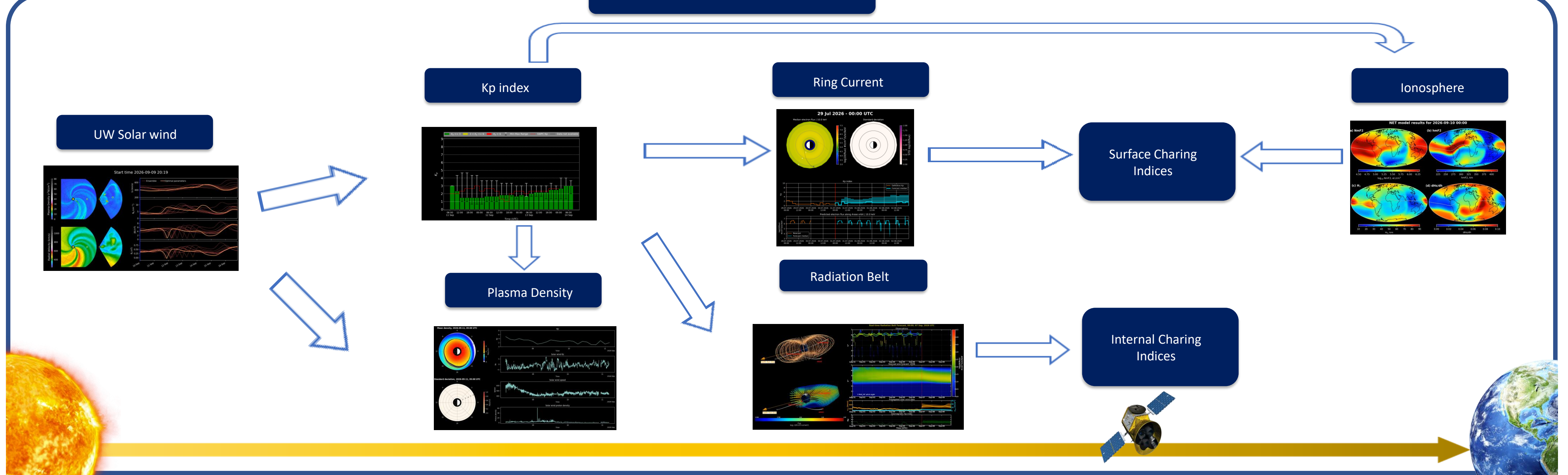
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1. GFZ, 2. ESA, 3. University of Warwick, 4. UCLouvain, 5. BIRA, 6. TRAD

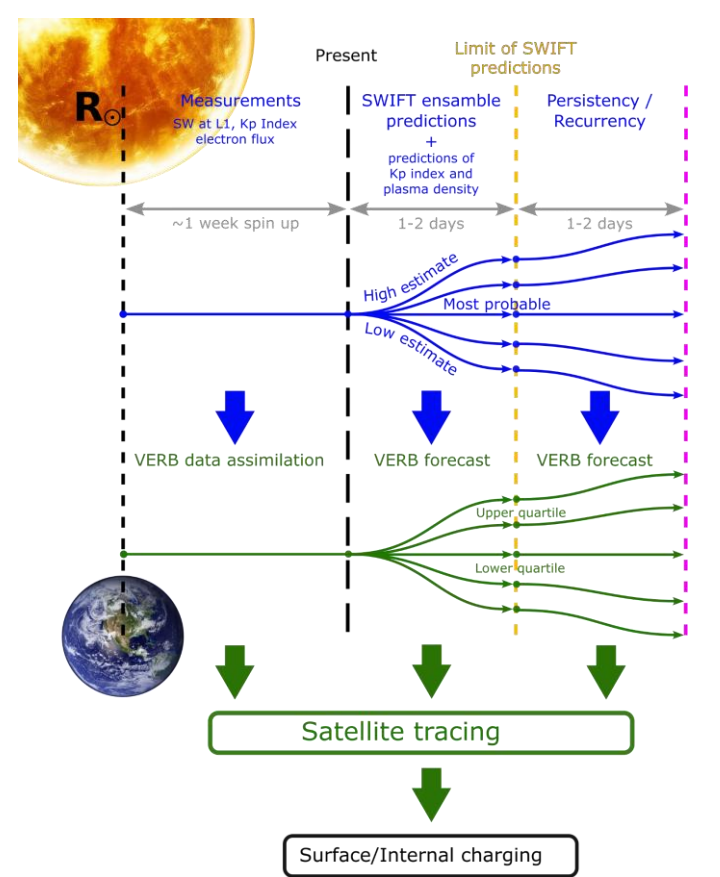
Highlights

1. Probabilistic space-weather forecasting from the Sun to Ionosphere, providing predictions together with information on forecast uncertainty.
2. Integration of physics-based models, machine learning and data assimilation to represent the different components of the space-weather system.
3. Improved ensemble forecasting and uncertainty quantification to increase forecast reliability and provide confidence information for users.
4. New ionospheric and magnetospheric predictions, including global TEC, ion and electron density, and ring current ion predictions.
5. New surface and internal spacecraft charging indicators, integrated with the scientific models within a coupled forecasting framework.

Overview



Ensemble Solution



The simulations are divided into **three steps**:

1. Measured solar wind parameters and a quick-look Kp index are used for data assimilative simulations with the VERB code to produce a nowcast of the space environment and to provide accurate initial conditions for the prediction steps.
2. SWIFT ensemble predictions of the solar wind parameters at L1 to run ensemble VERB simulations 1–2 days into the future.
3. we extend the VERB ensemble forecast beyond SWIFT predictions, by using predictions of the solar wind and the Kp based on recurrence and persistence.

Fig 2. Schematic illustration of the ensemble VERB forecast of ring current and radiation belt electron fluxes.

Validation

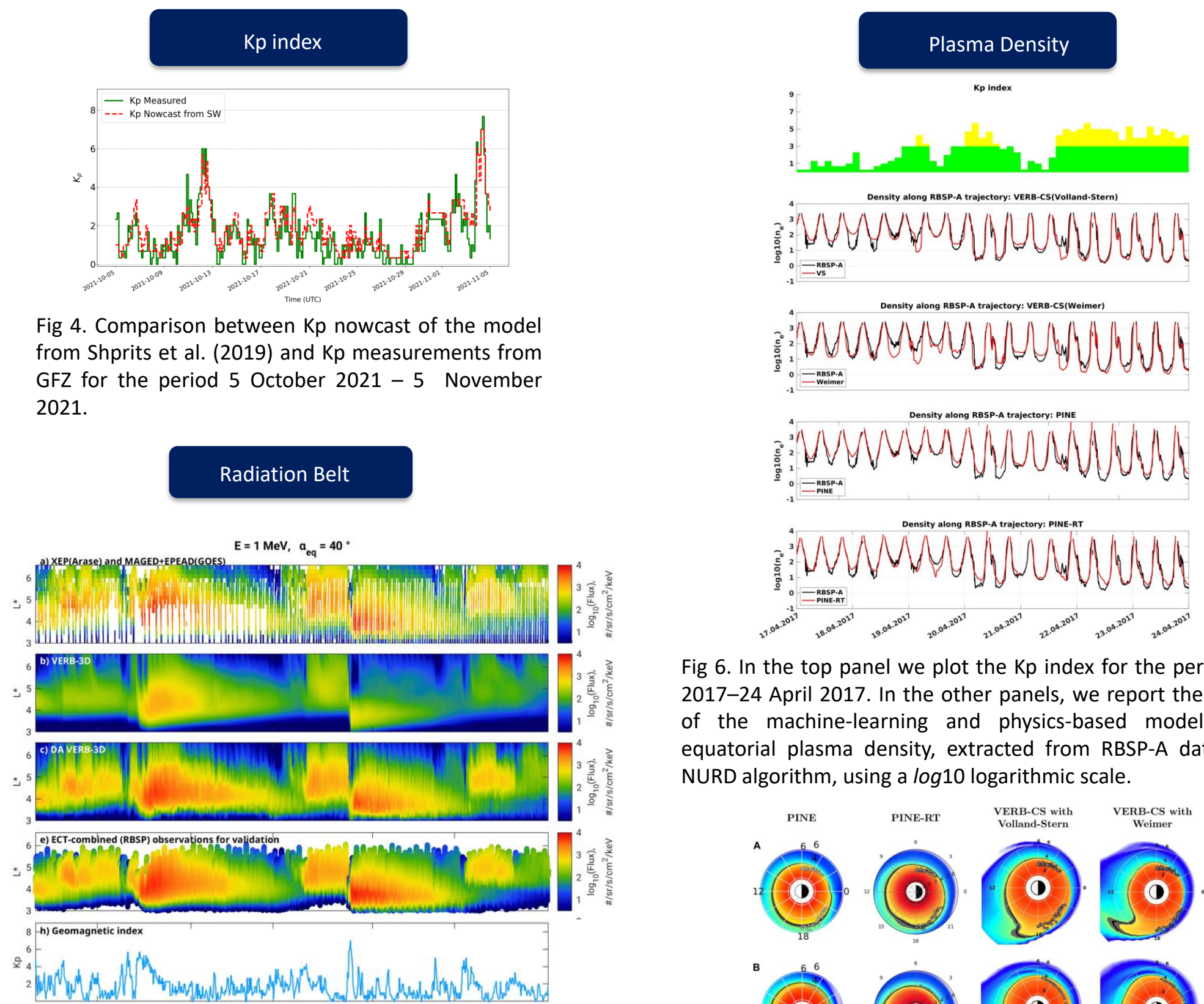


Fig 4. Comparison between Kp nowcast of the model from Shprits et al. (2019) and Kp measurements from GFZ for the period 5 October 2021 – 5 November 2021.

Fig 6. In the top panel we plot the Kp index for the period 17 April 2017–24 April 2017. In the other panels, we report the predictions of the machine-learning and physics-based models and the equatorial plasma density, extracted from RBSP-A data with the NURD algorithm, using a $\log_{10}$ logarithmic scale.

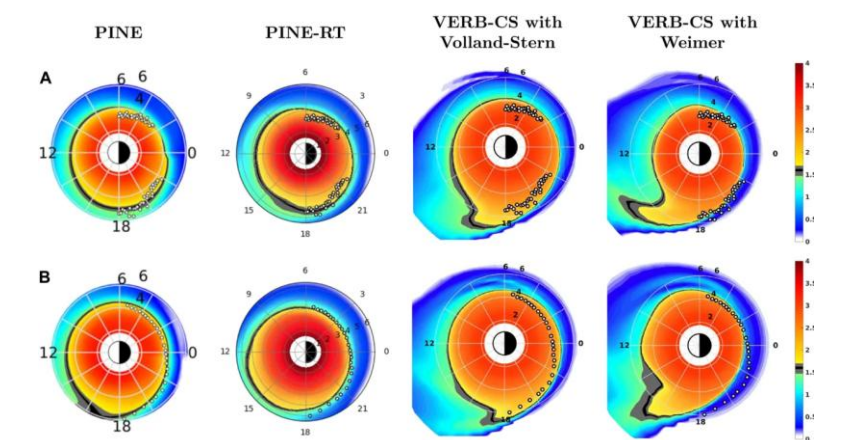


Fig 7. Comparison of the outputs of the PINE-RT model, of the PINE model and of the physics-based model driven by the Volland-Stern and by the Weimer electric field models. The color indicates the $\log_{10}$ of the plasma density (the scale of the colorbar is the same for all models and all times). The black-and-white dots show the location of the plasmopause derived from the IMAGE EUV images. The gray and black section of the colorbar indicates the density threshold of $40 \pm 10 \text{ el/cm}^3$, which is considered as an approximation of the plasmopause location. The Sun is to the left. Event (A): 11 Apr 2001 00:14 UT, Kp = 3. Event (B): 18 June 2001 03:45 UT, Kp = 4+.

Orchestration

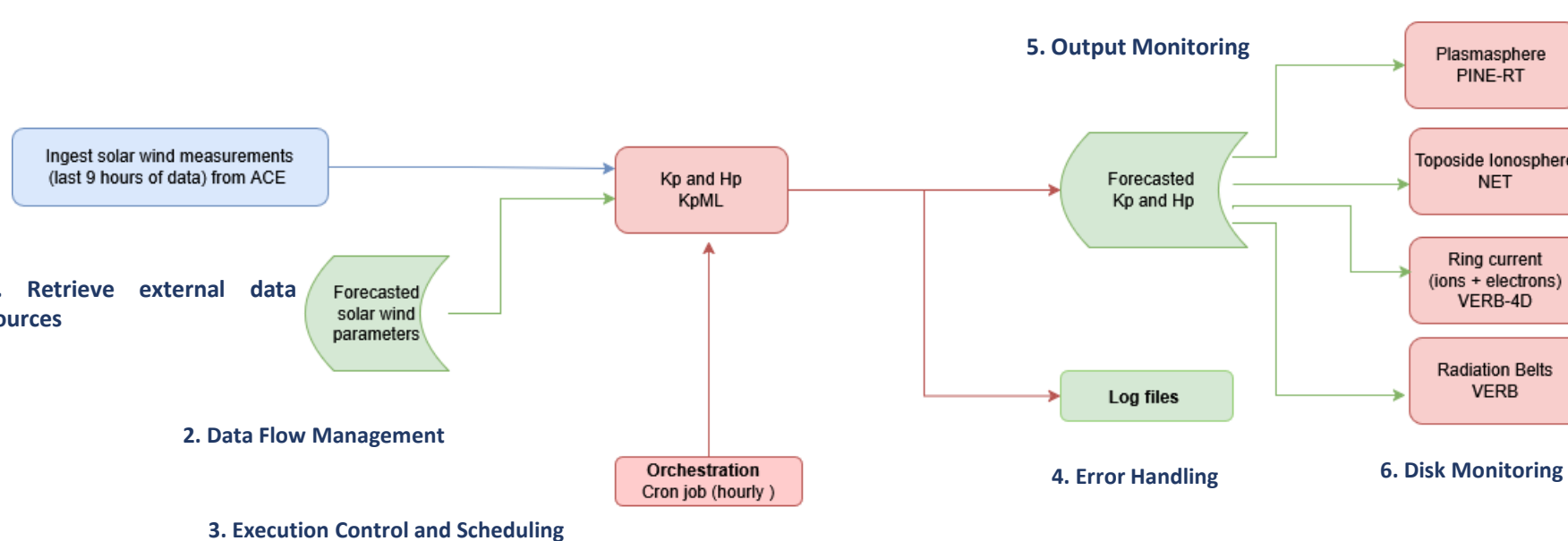


Fig 3. Schematic illustration of orchestration workflow.

1. Heterogeneous Models:

Different physics domains, codebases, and computational requirements, no common execution standard out of the box

2. Variable Runtimes

Models span from minutes to hours of compute time, requiring careful scheduling to meet latency targets.

3. Inconsistent I/O Formats

Input and output schemas differ across models, formats, units, temporal resolution, and file conventions.

4. Strict Dependencies

Downstream models depend on outputs of upstream ones, a failure anywhere in the chain must be detected and handled gracefully.

References

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Conclusion

1. FLAG provides an integrated probabilistic framework for forecasting solar wind, magnetospheric and ionospheric conditions and their potential effects on spacecraft.
2. The combination of ensemble predictions, uncertainty quantification and multiple modelling approaches enables forecasts to be accompanied by information on their reliability.
3. The addition of ionospheric, ring current and spacecraft charging products broadens the range of environmental conditions and space-weather effects covered by the forecasting system.
4. Ongoing work focuses on completing the surface and internal charging models, coupling the remaining components into the forecasting framework, and performing comprehensive verification and validation.
5. Through these developments, FLAG aims to integrate advanced scientific models into a reliable and usable prototype forecasting framework for the European Commission and its EU Space Programme.